

**11.12 Classwork: Exponential Functions and Compound Interest (calculator) —
Markscheme**

1. [Maximum mark: 6]

The value of a car is given by the function $C = 40\,000(0.91)^t$, where t is in years since 1 January 2023 and C is in USD (\$).

- (a) Write down the annual rate of depreciation of the car. [1]

$$1 - 0.91 = 0.09 = 9\% \quad (\mathbf{A1})$$

- (b) Find the value of the car on 1 January 2028. [2]

$$t = 5: C = 40\,000(0.91)^5 \quad (\mathbf{M1})$$

$$= 40\,000 \times 0.62403 \dots = 24\,961.29 \dots \approx \$24\,961 \text{ (3 s.f.)} \quad (\mathbf{A1})$$

- (c) Find the value of M . [3]

Alvie's account after 5 years:

$$A = 15\,000(1.03)^5 = 15\,000 \times 1.15927 \dots = 17\,389.11 \dots \quad (\mathbf{M1})(\mathbf{A1})$$

$$M = 24\,961.29 \dots - 17\,389.11 \dots = 7572.17 \dots \approx \$7570 \text{ (3 s.f.)} \quad (\mathbf{A1})$$

Follow through from parts (a) and (b).

2. [Maximum mark: 16]

Two friends Amelia and Bill, each set themselves a target of saving \$20 000. They each have \$9000 to invest.

(a) Amelia invests her \$9000 at 7% per annum compounded annually.

$$(i) A = 9000(1.07)^5 = 9000 \times 1.40255 \dots \quad (\text{M1})$$

$$= 12\,622.97 \dots \approx \$12\,600 \quad (\text{A1})$$

$$(ii) 9000(1.07)^n = 20\,000 \quad (\text{M1})$$

$$(1.07)^n = \frac{20\,000}{9000} = 2.2222 \dots$$

$$n = \frac{\ln 2.2222 \dots}{\ln 1.07} = \frac{0.7985 \dots}{0.06766 \dots} = 11.80 \dots \quad (\text{A1})$$

$$n = 12 \text{ years} \quad (\text{A1})$$

*Award **(M1)** for setting up the equation. Final answer must be a whole number (round up).*

(b) Bill invests \$9000 at $r\%$ compounded monthly. [3]

$$9000 \left(1 + \frac{r}{1200}\right)^{120} = 20\,000 \quad (\text{M1})$$

$$\left(1 + \frac{r}{1200}\right)^{120} = \frac{20\,000}{9000} = 2.2222 \dots$$

$$1 + \frac{r}{1200} = (2.2222 \dots)^{1/120} = 1.006681 \dots \quad (\text{A1})$$

$$r = 1200 \times 0.006681 \dots = 8.02 \dots$$

$$r = 8.02 \text{ (to 2 d.p.)} \quad (\text{A1})$$

Accept 8.02. Must be to 2 decimal places as stated.

(c) Chris: deposits a initially, then $\frac{a}{2}$, $\frac{a}{4}$, $\frac{a}{8}$, $\dots$ each year. No interest.

$$(i) \text{ Geometric series with } u_1 = 9000, r = \frac{1}{2}. \quad (\text{M1})$$

$$S_\infty = \frac{9000}{1 - \frac{1}{2}} = \frac{9000}{\frac{1}{2}} = 18\,000 \quad (\text{M1})(\text{A1})$$

Since $S_\infty = 18\,000 < 20\,000$, Chris can never reach the target. **(R1)(AG)**

$$(ii) \text{ After 5 years (deposits at } t = 0, 1, 2, 3, 4), \text{ total is:} \quad (\text{M1})$$

$$S_5 = \frac{a(1 - (0.5)^5)}{1 - 0.5} = \frac{a(1 - 0.03125)}{0.5} = \frac{0.96875a}{0.5} = 1.9375a \quad (\text{M1})(\text{A1})$$

$$1.9375a = 20\,000 \implies a = \frac{20\,000}{1.9375} = 10\,322.58\dots$$

$$a = \$10\,323 \text{ (to nearest dollar)} \quad \textbf{(A1)}$$

Must round up to ensure target is reached.